

Jan 18, 2021

Introduction to Computer and Network Security

Part 1

Question 1 [35 points]

Define the notion of hash function and its main properties: (a) Ease of computation, (b) Compression, (c) One-way, (d) Weak collision resistance, and (e) Strong collision resistance.

Question 2 [15 points]

What is a digital certificate and what are its main components?

Part 2

Question 3 [35 points]

Explain the notion of Multi-level security by illustrating it with the Bell-La Padula model; in particular discuss the notions of (a) security level, (b) need-to-know, (c) dominance relation, (d) no-read-up rule, (e) no-write-down rule.

Question 4 [15 points]

Explain the notion of pseudo anonymization function, describe at least two possible implementations together with their advantages and disadvantages.

Please, remember...

On each sheet, make sure to write: Name Surname Student ID number Question number	The file containing the scan of your answers should be named as follows: surname-name-studentIDnumber.pdf
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